Barrier Methods

The methods in Chapters 15 and 16 construct subproblems that help approach a constrained KKT point, usually without preserving primal feasibility at every iterate. We now ask whether an algorithm can instead keep every inequality strictly satisfied. To develop such a method, we will place a logarithmic wall at each inequality boundary, examine how its weight moves a subproblem minimizer, and derive the associated multiplier and complementarity equations. The resulting outer iteration lowers the walls, while an inner solver must make progress without crossing them.

Learning objectives

By the end of this chapter, you should be able to:

- identify the strict interior and form a logarithmic-barrier subproblem;
- solve a scalar barrier stationarity equation and select its interior branch;
- derive the induced multiplier, perturbed complementarity, and limiting KKT pair;
- construct Newton steps and protect strict inequality feasibility;
- distinguish inner stationarity, outer complementarity, and parameter accuracy; and
- compare barrier methods with QP, ALM, and SQP.

17.1 From primal-infeasible methods to barrier methods

Recall first the equality-constrained constructions. The quadratic penalty (QP) method minimizes $f + \rho \|\boldsymbol{g}\|^2/2$, whereas the augmented Lagrangian method (ALM) minimizes $f + (\boldsymbol{\lambda}^{(k)})^{\mathsf{T}}\boldsymbol{g} + \rho \|\boldsymbol{g}\|^2/2$. Their inner problems are unconstrained. Equality SQP instead minimizes a quadratic model subject to $\boldsymbol{g}(\boldsymbol{x}^{(k)}) + \boldsymbol{A}(\boldsymbol{x}^{(k)})^{\mathsf{T}}\boldsymbol{d} = \boldsymbol{0}$. Its inner problem is a quadratic program with linear equality constraints.

None of these constructions generally forces a finite iterate to satisfy the original constraints exactly. For QP and ALM the constraints appear in the objective; for SQP it is their linearization that is imposed. Even from an exactly feasible point, nonlinear constraints need not hold after an SQP step. Their general-inequality extensions likewise need not preserve inequality feasibility. In this sense they are *primal-infeasible methods*: the term describes the absence of a feasibility guarantee, rather than a claim that every iterate must be infeasible.

The two-level framework remains useful. An outer iteration forms a subproblem, and an inner iteration approximately solves it. We still use the original KKT residual from Chapter 15 and the target $r_{\mathrm{KKT}} \le \varepsilon_{\mathrm{out}}$; the inner tolerance is $\varepsilon_{\mathrm{in}}^{(k)}$. The new design requirement is that the solver preserves strict inequality feasibility throughout its accepted iterations. Equality

feasibility will still be handled by the inner solver.

17.1.1 The strict domain and the logarithmic wall

Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open and consider

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathcal{D}} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}, \\ & h_j(\boldsymbol{x}) \le 0, \qquad j = 1, \dots, p. \end{aligned} \tag{17.1}$$

Here $p \ge 1$. Assume the functions are twice continuously differentiable wherever Newton derivatives are used. Write

$$\mathcal{D}_< = \{\boldsymbol{x} \in \mathcal{D} : \boldsymbol{h}(\boldsymbol{x}) < \boldsymbol{0}\}, \qquad \mathcal{S}^\circ = \{\boldsymbol{x} \in \mathcal{D}_< : \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}\}.$$

Here $\mathcal{S}^\circ$ denotes strict inequality feasibility on the equality set; it need not be the topological interior of the full feasible set in $\mathbb{R}^n$. The basic barrier construction of this chapter requires $\mathcal{S}^\circ \neq \emptyset$. For example, the feasible set of $x \le 0$, $-x \le 0$ is $\{0\}$, but its strict interior is empty. Replacing these two inequalities by the equality $x = 0$ gives a more suitable representation.

For a slack $s > 0$, the scalar function $-\log s$ tends to $+\infty$ as $s \downarrow 0$. Substituting $s_j = -h_j(\boldsymbol{x})$ and assigning a positive weight μ therefore gives the objective

$$\Phi_\mu(\boldsymbol{x}) = f(\boldsymbol{x}) - \mu \sum_{j=1}^{p} \log(-h_j(\boldsymbol{x})), \qquad \boldsymbol{x} \in \mathcal{D}_<. \tag{17.2}$$

For each fixed $\mu > 0$, we solve the *barrier subproblem*

$$\min_{\boldsymbol{x}\in\mathcal{D}_<} \Phi_\mu(\boldsymbol{x}) \qquad \text{subject to} \qquad \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}.$$

The equalities remain explicit: $g_i = 0$ has no distinguished interior side, so a term such as $\log(-g_i)$ would describe a different problem. Near a finite boundary point at which the remaining terms stay finite, the relevant barrier term diverges. This discourages a minimizer from lying on that boundary, but an algorithm must still reject trial points outside $\mathcal{D}_<$ before evaluating the logarithms. Nor does the barrier alone guarantee that a minimizer exists; an unbounded domain or objective may require additional assumptions.

To check whether the added wall preserves convexity, suppose that the underlying domain is convex and each h_j is convex there. Use the composition rule: $\varphi(t) = -\log(-t)$ is convex and increasing for $t < 0$, since $\varphi'(t) = -1/t > 0$ and $\varphi''(t) = 1/t^2 > 0$. Indeed, for $0 \le \theta \le 1$,

$$\begin{aligned} \varphi(h_j(\theta\boldsymbol{x} + (1-\theta)\boldsymbol{y})) &\le \varphi(\theta h_j(\boldsymbol{x}) + (1-\theta)h_j(\boldsymbol{y})) \\ &\le \theta\varphi(h_j(\boldsymbol{x})) + (1-\theta)\varphi(h_j(\boldsymbol{y})). \end{aligned}$$

Thus each barrier term is convex when its inequality function is convex. If f is also convex, adding the terms proves convexity of Φ_μ; strict convexity of f implies strict convexity of Φ_μ. With affine equalities, the convex optimality and uniqueness results from Chapter 14 apply whenever a minimizer exists.

The logarithm is not the only function that can create a wall. For example, $-1/h_j$ also diverges to $+\infty$ as $h_j \uparrow 0$. Its scalar outer function is increasing and convex on $(-\infty, 0)$, so the same composition argument applies to a convex h_j. We will use the negative logarithm throughout because its derivative leads directly to a simple perturbed complementarity equation.

17.1.2 A one-dimensional example

To see the construction explicitly, consider

$$\min_{x\in\mathbb{R}} f(x) = (x-2)^2 \qquad \text{subject to} \qquad h(x) = x - 1 \le 0. \tag{17.3}$$

The unconstrained minimizer is 2, so the constrained minimizer is $x^* = 1$. Original KKT stationarity gives $2(x^* - 2) + \nu^* = 0$, hence $\nu^* = 2$. For $\mu > 0$, the barrier subproblem is

$$\min_{x<1} \Phi_\mu(x) = (x-2)^2 - \mu \log(1-x).$$

Its landscape with a varying μ is shown in Figure 17.1.

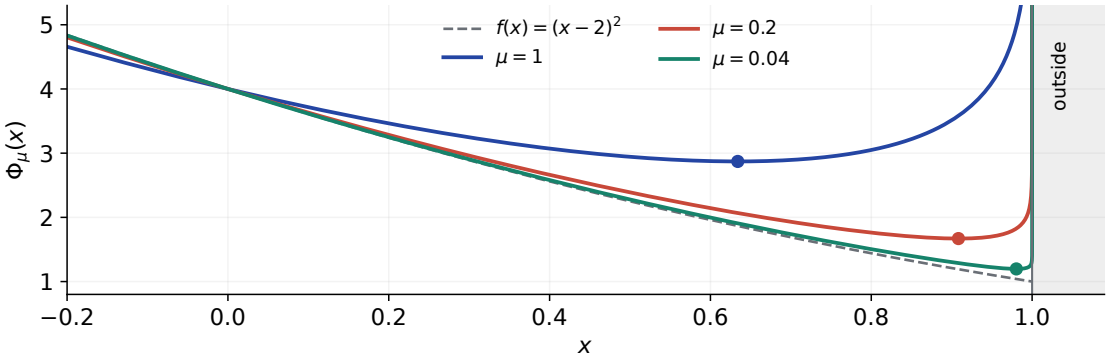


Figure 17.1: The scalar barrier objective for several positive values of μ. The logarithmic wall excludes the boundary $x = 1$, but its contribution is negative when $x < 0$. Decreasing its weight moves the minimizer toward the constrained solution from the strict side.

We now compute its minimizer rather than assume it is near 1. Differentiation gives

$$\Phi'_\mu(x) = 2(x-2) + \frac{\mu}{1-x}, \qquad \Phi''_\mu(x) = 2 + \frac{\mu}{(1-x)^2} > 0.$$

Thus any stationary point in $(-\infty, 1)$ is the unique global minimizer. Set $s = 1 - x > 0$. The stationarity equation becomes

$$-2(1+s) + \frac{\mu}{s} = 0 \quad \Longleftrightarrow \quad 2s^2 + 2s - \mu = 0.$$

The quadratic formula has two roots. The negative root violates $s > 0$, so we retain

$$s(\mu) = \frac{\sqrt{1+2\mu} - 1}{2} = \frac{\mu}{1 + \sqrt{1+2\mu}}, \qquad x(\mu) = 1 - s(\mu). \tag{17.4}$$

The rationalized expression for s avoids subtracting two nearly equal numbers when μ is small. Because $s(\mu) > 0$, every point on this path is strictly feasible. As $\mu \downarrow 0$, $s(\mu) \downarrow 0$ and $x(\mu) \uparrow 1$.

Next compare barrier stationarity with original KKT stationarity. This identifies

$$\nu(\mu) = \frac{\mu}{1 - x(\mu)} = 1 + \sqrt{1+2\mu} = 2(1 + s(\mu)). \tag{17.5}$$

Consequently $\nu(\mu) > 0$, $\nu(\mu)(1 - x(\mu)) = \mu$, and $\nu(\mu) \to 2$. The limiting pair $(1, 2)$ satisfies all the original KKT conditions. This exact family of barrier minimizers is a *central path*;

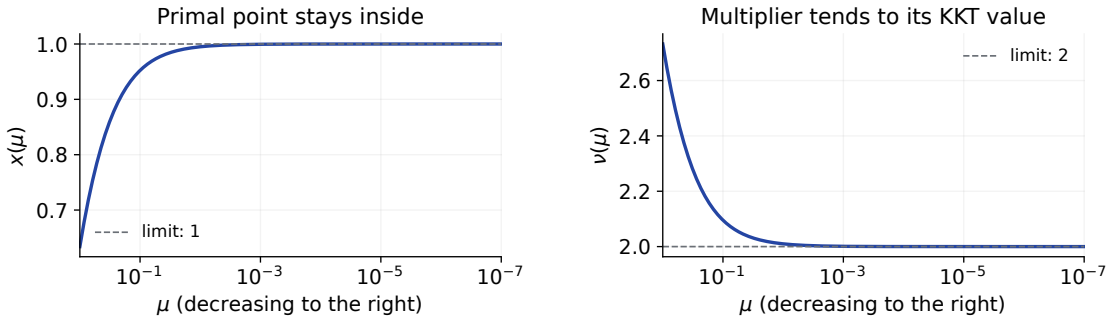


Figure 17.2: The exact scalar barrier path and its induced multiplier. The formulas show $x(\mu) \uparrow 1$ and $\nu(\mu) \to 2$ as $\mu \downarrow 0$, independently of any inner solver.

the points returned by an inexact numerical method generally only approximate it. Notice the order of the calculation: the barrier domain selects the root, stationarity identifies the multiplier, and only then do we pass to the limit.

For orientation, the following values are obtained from the exact formulas; they are not iteration counts or measurements of an inner solver.

μ	$x(\mu)$	$s(\mu)$	$\nu(\mu)$	$\nu(\mu)s(\mu)$
1.0000	0.6339746	0.3660254	2.732051	1.0000
0.2000	0.9083920	0.0916080	2.183216	0.2000
0.0400	0.9803848	0.0196152	2.039230	0.0400
0.0080	0.9960159	0.0039841	2.007968	0.0080
0.0016	0.9992006	0.0007994	2.001599	0.0016

17.1.3 An idealized barrier method

Before addressing implementation, suppose that each chosen barrier subproblem has a minimizer and a corresponding equality multiplier. We can then express the outer idea without specifying an inner algorithm.

Algorithm: Naive barrier method: exact solves

Choose $\mu^{(0)} > 0$ and a fixed $0 < \theta < 1$. For $k = 0, 1, \ldots$:

1. Solve the barrier subproblem exactly:

$$\boldsymbol{x}^{(k+1)} \in \operatorname*{arg\,min}_{\boldsymbol{x} \in \mathcal{D}_<,\ \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}} \Phi_{\mu^{(k)}}(\boldsymbol{x}).$$

   Obtain a corresponding equality multiplier $\boldsymbol{\lambda}^{(k+1)}$.

2. Form $\nu_j^{(k+1)} = \mu^{(k)} / [-h_j(\boldsymbol{x}^{(k+1)})]$ and inspect the original KKT residual.

3. If the required accuracy is not reached, set $\mu^{(k+1)} = \theta \mu^{(k)}$ and solve the next subproblem.

This idealization explains the parameter sequence, but leaves three practical questions unanswered: how to obtain an interior start, how accurately to solve a subproblem, and how

to keep an inner step inside its domain. Nonconvex inner solvers also generally find local stationary points rather than global minimizers. We first explain why the barrier parameter should be driven to zero, then return to these implementation questions.

17.1.4 Why the barrier parameter tends to zero

First consider a fixed point in the strict domain. Each added term $-\mu \log(-h_j(\boldsymbol{x}))$ tends to zero as $\mu \downarrow 0$, so the barrier objective approaches f there. For a finite positive μ, however, the wall changes the objective throughout the strict domain. In particular,

$$-\mu \log(-h_j) \begin{cases} > 0, & -1 < h_j < 0, \\ = 0, & h_j = -1, \\ < 0, & h_j < -1. \end{cases}$$

The word "barrier" therefore does not mean a nonnegative penalty: far enough inside, this term lowers the original objective; see the left panel of Figure 17.3. Reducing its weight removes this interior bias pointwise. The convergence is not uniform near a boundary, where the logarithm still diverges for every positive μ.

We next examine the optimality equations to see why decreasing μ is also required by complementarity. Repeat the scalar calculation for (17.1). Define

$$\boldsymbol{A}(\boldsymbol{x}) = [\nabla g_1(\boldsymbol{x}), \dots, \nabla g_m(\boldsymbol{x})], \qquad \boldsymbol{C}(\boldsymbol{x}) = [\nabla h_1(\boldsymbol{x}), \dots, \nabla h_p(\boldsymbol{x})].$$

For one barrier term, the chain rule gives

$$\nabla[-\mu \log(-h_j)] = -\frac{\mu}{h_j} \nabla h_j.$$

Thus, if we introduce the positive quantities

$$\nu_j(\boldsymbol{x}, \mu) = \frac{\mu}{-h_j(\boldsymbol{x})} > 0, \tag{17.6}$$

then

$$\nabla \Phi_\mu(\boldsymbol{x}) = \nabla f(\boldsymbol{x}) + \sum_{j=1}^{p} \nu_j(\boldsymbol{x}, \mu) \nabla h_j(\boldsymbol{x}). \tag{17.7}$$

Let $(\boldsymbol{x}, \boldsymbol{\lambda})$ satisfy the KKT equations for a barrier subproblem. Its stationarity condition is $\nabla \Phi_\mu(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})\boldsymbol{\lambda} = \boldsymbol{0}$. Substituting (17.7) and defining $\boldsymbol{\nu}$ by (17.6), we obtain

$$\begin{aligned} \nabla f(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})\boldsymbol{\lambda} + \boldsymbol{C}(\boldsymbol{x})\boldsymbol{\nu} &= \boldsymbol{0}, \\ \boldsymbol{g}(\boldsymbol{x}) &= \boldsymbol{0}, \\ \boldsymbol{h}(\boldsymbol{x}) < \boldsymbol{0}, \qquad \boldsymbol{\nu} &> \boldsymbol{0}, \\ -\nu_j h_j(\boldsymbol{x}) &= \mu, \qquad j = 1, \dots, p. \end{aligned} \tag{17.8}$$

Compare this system with the original KKT conditions. Stationarity and equality feasibility are unchanged, primal and dual inequality feasibility are strict, and complementarity is replaced by a positive product. A suitable continuous family $(\boldsymbol{x}(\mu), \boldsymbol{\lambda}(\mu), \boldsymbol{\nu}(\mu))$ of these points is called a *central path*. This terminology does not imply that a path exists or is unique for every nonlinear problem; local sufficient assumptions are given in Section 17.5.

At any strict interior point, even before inner stationarity is reached, the induced multipliers satisfy

$$-\nu_j h_j = \mu, \qquad -\boldsymbol{\nu}^\mathsf{T}\boldsymbol{h} = p\mu.$$

If there is at least one inequality, a fixed positive μ therefore keeps the complementarity residual of these induced multipliers positive, even when the subproblem is solved exactly. To make the full original KKT residual vanish with these multipliers, we must let $\mu \to 0$. The sum is an aggregate complementarity measure. To turn it into an objective-error bound, we need convexity as well as exact barrier stationarity.

Lemma 17.1 (Convex barrier objective-error bound)**.** *Suppose $\mathcal{D}$ is convex, $f, h_1, \ldots, h_p$ are convex on $\mathcal{D}$, and the equalities are affine. Let $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu})$ satisfy the barrier KKT equations (17.8) exactly for $\mu > 0$, and let f^* denote the infimum of the original problem. Then $0 \le f(\boldsymbol{x}) - f^* \le p\mu$.*

Proof. For any feasible $\boldsymbol{y}$, convexity and stationarity give

$$\begin{aligned} f(\boldsymbol{y}) &\ge f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{x}) \\ &= f(\boldsymbol{x}) - \sum_j \nu_j \nabla h_j(\boldsymbol{x})^\mathsf{T}(\boldsymbol{y} - \boldsymbol{x}) \\ &\ge f(\boldsymbol{x}) + \sum_j \nu_j (h_j(\boldsymbol{x}) - h_j(\boldsymbol{y})) \\ &\ge f(\boldsymbol{x}) + \sum_j \nu_j h_j(\boldsymbol{x}) = f(\boldsymbol{x}) - p\mu. \end{aligned}$$

The equality-multiplier term vanishes because both points satisfy the affine equalities. Taking the infimum over feasible $\boldsymbol{y}$ gives $f^* \ge f(\boldsymbol{x}) - p\mu$; when a global minimizer exists, we can instead substitute it for $\boldsymbol{y}$. Since $\boldsymbol{x}$ is itself feasible, $f^* \le f(\boldsymbol{x})$, proving the stated bound. □

If $\mu^{(k)} \to 0$ and the full triples $(\boldsymbol{x}^{(k)}, \boldsymbol{\lambda}^{(k)}, \boldsymbol{\nu}^{(k)})$ converge while their stationarity and equality residuals tend to zero, continuity lets us pass to the limit in (17.8). The limiting triple satisfies original KKT. Convergence of the primal points alone does not automatically give convergence of the multipliers. Under LICQ at the primal limit, the active-gradient argument in Section 17.5 supplies that missing step. A nonconvex KKT limit still needs the second-order tests from Chapter 13; convexity gives the sufficiency result from Chapter 14.

17.1.5 Comparing barrier with penalty

To compare the two landscape modifications, consider the inequality part of QP:

$$P_\rho(\boldsymbol{x}) = f(\boldsymbol{x}) + \frac{\rho}{2}\left\|[\boldsymbol{h}(\boldsymbol{x})]_+\right\|^2, \qquad \Phi_\mu(\boldsymbol{x}) = f(\boldsymbol{x}) - \mu \sum_j \log(-h_j(\boldsymbol{x})).$$

Both alter the objective to influence where subproblem minimizers lie. If there are no equalities, both are minimized without explicit equality constraints, although the barrier has the open domain $\mathcal{D}_<$. Their treatment of an inequality is different:

- QP leaves the objective unchanged at all feasible points, including the boundary, and raises it outside. Its finite-parameter minimizers can be infeasible. Increasing ρ raises the cost of violations.

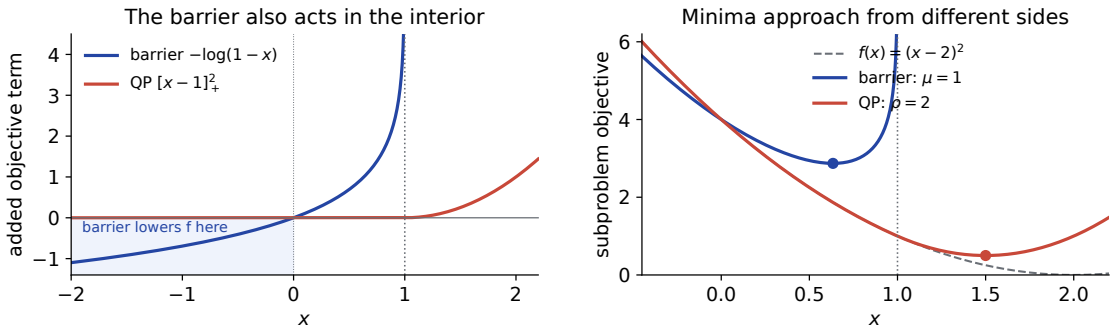


Figure 17.3: Logarithmic walls and quadratic exterior penalties for the scalar problem. The barrier acts throughout the strict domain and excludes its boundary; the inequality penalty is zero throughout the feasible set and grows outside. Their subproblem minimizers approach the solution from opposite sides.

- The barrier is defined only on the strict side and diverges at the boundary. It also changes values inside, sometimes lowering them. A numerical method must explicitly protect that domain. Decreasing μ reduces the interior bias and the complementary product of the induced multipliers.

See Figure 17.3 for an illustration. These descriptions concern the inequality terms; any equality penalties remain as specified by the respective method.

17.2 Initialization and inexact subproblem solves

17.2.1 Obtaining a starting point

When a point in $\mathcal{S}^\circ$ is not known, a conceptual Phase I problem is

$$\min_{\boldsymbol{x}\in\mathcal{D},t} t \qquad \text{subject to} \qquad \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}, \quad h_j(\boldsymbol{x}) \le t \quad (j = 1, \dots, p).$$

To check what this formulation achieves, first suppose $\boldsymbol{x}$ is strictly feasible for the original problem. Then $t = \max_j h_j(\boldsymbol{x}) < 0$ makes $(\boldsymbol{x}, t)$ Phase I feasible. Conversely, a Phase I feasible pair with $t < 0$ gives $h_j(\boldsymbol{x}) \le t < 0$ for every j. Thus strict feasibility is equivalent to finding a feasible negative value, or to a negative Phase I infimum (including $-\infty$). An algorithm may stop as soon as such a point is found or impose a lower bound on t; accurately minimizing an unbounded Phase I objective is unnecessary. Phase I itself still needs a suitable solver. With nonlinear equalities, equality restoration is a separate issue from finding positive inequality slacks.

For a problem without equalities, a simple numerical Phase I is to choose a margin $\eta > 0$ and reduce

$$R_\eta(\boldsymbol{x}) = \tfrac{1}{2} \left\| [\boldsymbol{h}(\boldsymbol{x}) + \eta \boldsymbol{1}]_+ \right\|^2, \qquad \nabla R_\eta(\boldsymbol{x}) = \boldsymbol{C}(\boldsymbol{x}) [\boldsymbol{h}(\boldsymbol{x}) + \eta \boldsymbol{1}]_+.$$

Use SD with Armijo backtracking, keep trials in $\mathcal{D}$, and stop once every $h_j(\boldsymbol{x}) \le -\eta/2$. This stopping condition supplies a strict interior point without requiring an exact minimizer of R_η. A fixed margin may be unattainable even when some strict point exists, and a local Phase I solver can fail; failure of this search is not a certificate that the original problem is

infeasible. A supplied point that is already strict can be used immediately without imposing this extra margin.

The function `find_strict_start` uses this construction with $\eta = 1/2$ and $h(\boldsymbol{x}) = x_1^2 + x_2 - 1$, the inequality of the common test below. Starting at $\boldsymbol{z}^{(0)} = (1, 2)$, the two accepted SD steps are

$$(1,2) \xrightarrow{\alpha=1/4} (-1/4, 11/8) \xrightarrow{\alpha=1} (7/32, 7/16), \qquad h(7/32, 7/16) = -527/1024 < -1/4;$$

see also Figure 17.4. These are initialization steps for R_η, not barrier optimization steps; the initial guess is allowed to be infeasible.

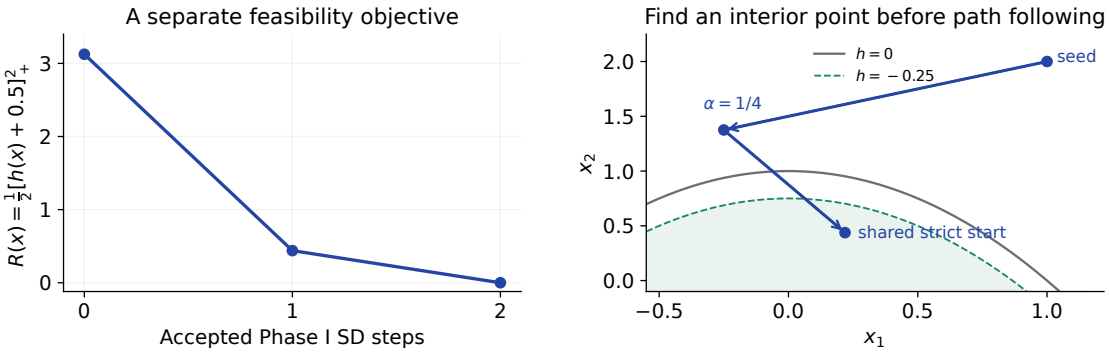


Figure 17.4: Phase I on the common test. Two accepted SD steps reduce the shifted-violation objective and produce a strict starting point. Interior initialization is a separate optimization task from minimizing the barrier objective.

17.2.2 Newton and equality-SQP directions

With no equalities, line-search steepest descent (SD) gives the direction $\boldsymbol{d} = -\nabla\Phi_\mu$; its directional derivative is $-\|\nabla\Phi_\mu\|^2 < 0$ away from stationarity. To use curvature information, Newton's method instead solves

$$\nabla^2\Phi_\mu(\boldsymbol{x})\boldsymbol{d} = -\nabla\Phi_\mu(\boldsymbol{x}).$$

Since $\nabla\nu_j = (\mu/h_j^2)\nabla h_j$, the product rule yields

$$\nabla^2\Phi_\mu(\boldsymbol{x}) = \nabla^2 f(\boldsymbol{x}) + \sum_{j=1}^{p} \nu_j \nabla^2 h_j(\boldsymbol{x}) + \sum_{j=1}^{p} \frac{\mu}{h_j(\boldsymbol{x})^2} \nabla h_j(\boldsymbol{x}) \nabla h_j(\boldsymbol{x})^\mathsf{T}. \tag{17.9}$$

The final sum is positive semidefinite, but the entire Hessian need not be positive definite in a nonconvex problem. When the Hessian is positive definite and the gradient is nonzero, multiplication by $\boldsymbol{d}^\mathsf{T}$ gives $\nabla\Phi_\mu^\mathsf{T}\boldsymbol{d} = -\boldsymbol{d}^\mathsf{T}\nabla^2\Phi_\mu\boldsymbol{d} < 0$. The direction is then a descent direction. An indefinite Hessian needs the same direction safeguards as an unconstrained Newton method; an arbitrary Newton root step is not automatically a minimization step.

Even a descent direction need not give an admissible full step. In the scalar example (17.3), take $x = 0$ and $\mu = 0.04$. Then

$$\Phi_\mu'(0) = -3.96, \qquad \Phi_\mu''(0) = 2.04, \qquad d = 3.96/2.04 \approx 1.941.$$

The full step lies beyond $x = 1$, outside the logarithm's domain. We will therefore impose a domain test as well as sufficient decrease when choosing a step length.

With equalities, introduce $L_\mu(\boldsymbol{x}, \boldsymbol{\lambda}) = \Phi_\mu(\boldsymbol{x}) + \boldsymbol{\lambda}^\mathsf{T}\boldsymbol{g}(\boldsymbol{x})$. Linearizing its stationarity equation and $\boldsymbol{g} = \boldsymbol{0}$ gives the Lagrange–Newton system from Chapter 16:

$$\begin{pmatrix} \nabla^2_{\boldsymbol{xx}} L_\mu & \boldsymbol{A} \\ \boldsymbol{A}^\mathsf{T} & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{d} \\ \boldsymbol{\delta} \end{pmatrix} = - \begin{pmatrix} \nabla \Phi_\mu + \boldsymbol{A}\boldsymbol{\lambda} \\ \boldsymbol{g} \end{pmatrix}. \tag{17.10}$$

In terms of the ordinary Lagrangian $L = f + \boldsymbol{\lambda}^\mathsf{T}\boldsymbol{g} + \boldsymbol{\nu}^\mathsf{T}\boldsymbol{h}$, its upper-left block is

$$\nabla^2_{\boldsymbol{xx}} L_\mu = \nabla^2_{\boldsymbol{xx}} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) + \boldsymbol{C}\,\mathrm{diag}\left(\frac{\mu}{h_j(\boldsymbol{x})^2}\right)\boldsymbol{C}^\mathsf{T}.$$

As in Chapter 16, a full LN step is a local root-finding step and need not produce descent from a distant point. The equality-SQP interpretation, curvature regularization, and merit-function safeguards from that chapter apply to the barrier objective as well. A line-searched SQP inner method is therefore a way to combine equality restoration with global progress toward a subproblem KKT point under the corresponding assumptions.

Before choosing a step length, distinguish equality feasibility from strict inequality feasibility. The second block equation enforces only $\boldsymbol{g} + \boldsymbol{A}^\mathsf{T}\boldsymbol{d} = \boldsymbol{0}$. If $\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}$ and $\boldsymbol{g}$ is nonlinear, Taylor expansion gives $\boldsymbol{g}(\boldsymbol{x}+\alpha\boldsymbol{d}) = O(\alpha^2\,\|\boldsymbol{d}\|^2)$, which need not vanish. For affine equalities, a tangent step from an equality-feasible point does preserve them. Thus the inner method may start from any $\boldsymbol{x} \in \mathcal{D}_<$ when it handles equality infeasibility; its stopping test must still check $\boldsymbol{g}$.

17.2.3 Protecting the domain of the logarithm

We first derive a step bound for affine inequalities, for which the linear prediction is exact.

Lemma 17.2 (Affine fraction-to-boundary rule)**.** *Let $h_j(\boldsymbol{x}) = \boldsymbol{a}_j^\mathsf{T}\boldsymbol{x} - b_j$, and suppose $h_j(\boldsymbol{x}) < 0$ for every j. For a direction $\boldsymbol{d}$ and $0 < \tau < 1$, define*

$$\alpha_{\max} = \min\left\{1,\ \tau \min_{\boldsymbol{a}_j^\mathsf{T}\boldsymbol{d} > 0} \frac{-h_j(\boldsymbol{x})}{\boldsymbol{a}_j^\mathsf{T}\boldsymbol{d}}\right\}. \tag{17.11}$$

The inner minimum is $+\infty$ if no constraint moves toward its boundary. Then $h_j(\boldsymbol{x}+\alpha\boldsymbol{d}) < 0$ for every j and every $0 \le \alpha \le \alpha_{\max}$.

Proof. Along the search direction,

$$h_j(\boldsymbol{x} + \alpha\boldsymbol{d}) = h_j(\boldsymbol{x}) + \alpha\boldsymbol{a}_j^\mathsf{T}\boldsymbol{d}.$$

If $\boldsymbol{a}_j^\mathsf{T}\boldsymbol{d} \le 0$, increasing $\alpha \ge 0$ does not move this constraint toward its boundary. If $\boldsymbol{a}_j^\mathsf{T}\boldsymbol{d} > 0$, its first boundary crossing occurs at $\alpha_j = -h_j(\boldsymbol{x})/(\boldsymbol{a}_j^\mathsf{T}\boldsymbol{d})$. For each constraint with positive directional derivative and $0 < \alpha \le \alpha_{\max}$,

$$h_j(\boldsymbol{x} + \alpha\boldsymbol{d}) \le h_j(\boldsymbol{x}) + \tau[-h_j(\boldsymbol{x})] = (1-\tau)h_j(\boldsymbol{x}) < 0.$$

The other constraints remain strict by their nonpositive directional derivatives. The case $\alpha = 0$ is the assumed strict feasibility of the current point. □

We can therefore start the line search at $\alpha_{\max}$ for affine inequalities; any separate domain restriction $\boldsymbol{x} + \alpha\boldsymbol{d} \in \mathcal{D}$ must also be respected.

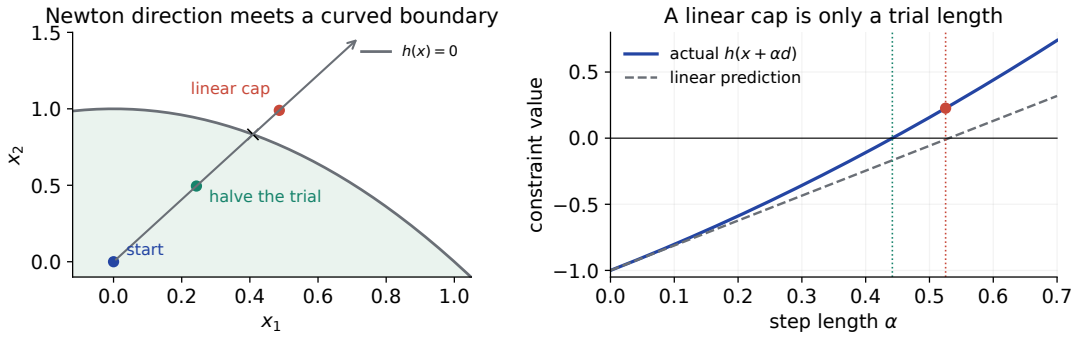


Figure 17.5: A Newton direction for a nonlinear inequality. The linear prediction overestimates the admissible step, so the actual constraint is tested before a logarithm is evaluated.

For a nonlinear h_j, replacing $\boldsymbol{a}_j$ by $\nabla h_j(\boldsymbol{x})$ in (17.11) gives only a linear prediction. The implementation must evaluate the actual $h_j(\boldsymbol{x} + \alpha \boldsymbol{d})$ and backtrack if any inequality fails. This feasibility backtracking terminates for a fixed finite direction: $\mathcal{D}$ is open, each h_j is continuous, and each current slack is strictly positive. These facts give a positive neighborhood of $\alpha = 0$ in which all the inequalities remain strict. Taking the minimum of the finitely many neighborhood radii gives a common admissible interval.

To see what can go wrong for an actual Newton direction, consider

$$f(\boldsymbol{x}) = \tfrac{1}{2}[(x_1 - 1)^2 + (x_2 - 2)^2], \qquad h(\boldsymbol{x}) = x_1^2 + x_2 - 1,$$

at $\boldsymbol{x} = (0, 0)$ with $\mu = 0.04$. The gradient and Hessian give

$$\nabla \Phi_\mu = (-1, -1.96)^{\mathsf{T}}, \qquad \nabla^2 \Phi_\mu = \operatorname{diag}(1.08, 1.04), \qquad \boldsymbol{d} \approx (0.925926, 1.884615)^{\mathsf{T}}.$$

The linear prediction reaches the boundary at $\alpha_{\text{lin}} = 1/1.884615 \approx 0.530612$. Along the actual step, however,

$$h(\alpha \boldsymbol{d}) = \alpha^2 d_1^2 + \alpha d_2 - 1,$$

whose positive root is approximately 0.441813. Even the predicted fraction-to-boundary cap $0.99\alpha_{\text{lin}} \approx 0.525306$ is infeasible. Halving it to approximately 0.262653 restores strict feasibility. We must still check objective decrease before accepting that trial. For illustration, refer to Figure 17.5.

17.2.4 Requiring progress as well as feasibility

Once a trial point lies in $\mathcal{D}_<$, the no-equality case uses the Armijo condition

$$\Phi_\mu(\boldsymbol{x} + \alpha \boldsymbol{d}) \le \Phi_\mu(\boldsymbol{x}) + \sigma \alpha \nabla \Phi_\mu(\boldsymbol{x})^{\mathsf{T}} \boldsymbol{d}, \qquad 0 < \sigma < 1.$$

If this fails, replace α by $\beta\alpha$, $0 < \beta < 1$, and repeat the domain test before evaluating the new objective. With equalities, use the equality merit function from Chapter 16, for example

$$M_{\mu,\omega}(\boldsymbol{x}) = \Phi_\mu(\boldsymbol{x}) + \omega \left\| \boldsymbol{g}(\boldsymbol{x}) \right\|_1 .$$

Here ω is an equality merit weight, not the barrier parameter. The equality-SQP descent condition determines a suitable weight and direction; strict inequality feasibility alone is insufficient for merit descent. Equality multipliers are updated consistently with the chosen equality solver.

Algorithm: A Newton inner solve without equality constraints

Fix $\mu > 0$, an inner tolerance $\varepsilon_{\text{in}}^{(k)}$, and a strict interior starting point. At each inner iterate:

1. Compute $\nabla\Phi_\mu$. Return the point if $\|\nabla\Phi_\mu\|_\infty \le \varepsilon_{\text{in}}^{(k)}$.
2. Solve $\nabla^2\Phi_\mu \boldsymbol{d} = -\nabla\Phi_\mu$. Check that the system was solved successfully and $\nabla\Phi_\mu^{\mathsf{T}}\boldsymbol{d} < 0$.
3. Start with the fraction-to-boundary prediction (17.11). First test the actual domain $\boldsymbol{x} + \alpha\boldsymbol{d} \in \mathcal{D}_<$; then test Armijo decrease. Reduce α by the fixed factor β until both tests pass.
4. Set $\boldsymbol{x} \leftarrow \boldsymbol{x} + \alpha\boldsymbol{d}$, and repeat.

If a direction, line search, or iteration limit fails, report an unsuccessful inner solve; do not reduce μ as though the subproblem had been solved.

The function `newton_subproblem` implements this procedure on the common test below. Its barrier Hessian is positive definite, so no curvature modification is needed. Applying the same procedure to a general nonconvex problem requires the direction safeguards already discussed.

17.2.5 Boundary curvature and numerical accuracy

To understand the cost of decreasing μ, inspect the final term in (17.9). Along a central path, if an active constraint has $\nu_j(\mu) \to \nu_j^* > 0$, perturbed complementarity gives

$$-h_j(\boldsymbol{x}(\mu)) = \frac{\mu}{\nu_j(\mu)} = \Theta(\mu), \qquad \frac{\mu}{h_j(\boldsymbol{x}(\mu))^2} = \frac{\nu_j(\mu)^2}{\mu} = \Theta(1/\mu).$$

Thus the scalar coefficient is $\Theta(1/\mu)$. If its limiting constraint gradient is nonzero, the corresponding rank-one matrix has the same order of magnitude. An inactive constraint with limiting positive slack instead contributes a coefficient $O(\mu)$. Large active-normal curvature together with moderate curvature in other directions can make a multivariate Hessian ill-conditioned. This is analogous to the normal-direction stiffness of QP as $\rho \uparrow \infty$.

In the scalar example, $\Phi_\mu''(x(\mu)) = 2 + \nu(\mu)^2/\mu$ grows like $4/\mu$. A nonzero 1×1 matrix nevertheless has condition number one, so this example by itself does not demonstrate ill-conditioning. It demonstrates a narrow region in which the Newton model is accurate: the third derivative $2\mu/(1-x)^3$ grows like $\Theta(1/\mu^2)$. Even with exact derivatives, a full Newton step can cross the boundary. Warm starts and backtracking address this nonlinearity; scaling and stable factorizations address the additional linear-algebra difficulties in larger problems.

These are two distinct difficulties. In exact arithmetic, Newton's method is unchanged by an invertible affine change of coordinates, so an ill-conditioned Hessian alone does not imply a slower exact Newton iteration. In floating-point arithmetic, badly scaled linear systems can lose accuracy. Separately, the rapid change in the barrier's derivatives can make the region of reliable quadratic approximation small, even when the Newton equations are solved exactly.

Near the boundary, also distinguish a small step from small stationarity. The ratio $\mu/(-h_j)$ is sensitive to errors in a very small slack. Compute slacks carefully, check the actual KKT residuals, and report a failed or stagnated inner solve instead of declaring convergence merely because an iterate no longer moves in floating-point arithmetic.

17.3 A practical barrier method

We can now combine initialization, the inner solve, and the parameter update. First relate the accuracy of a returned subproblem point to the original KKT conditions. At outer iteration k, the inner solver keeps $\mu^{(k)}$ fixed. Let its returned point be $(\boldsymbol{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)})$. To measure its inner accuracy, use

$$r_{\text{in}} = \max\left\{ \left\| \boldsymbol{g}(\boldsymbol{x}^{(k+1)}) \right\|_\infty, \left\| \nabla \Phi_{\mu^{(k)}}(\boldsymbol{x}^{(k+1)}) + \boldsymbol{A}(\boldsymbol{x}^{(k+1)})\boldsymbol{\lambda}^{(k+1)} \right\|_\infty \right\}.$$

Define $\nu_j^{(k+1)} = \mu^{(k)}/[-h_j(\boldsymbol{x}^{(k+1)})]$. By (17.7), the inner stationarity residual is exactly the original KKT stationarity residual for these multipliers. Strict inequality feasibility and positivity of $\boldsymbol{\nu}$ make their corresponding violation components zero. Moreover, $r_{\text{comp}} = \|\boldsymbol{\nu} \odot \boldsymbol{h}\|_\infty = \mu^{(k)}$. Hence, in exact arithmetic,

$$r_{\text{KKT}} = \max\{r_{\text{stat}}, r_{\text{feas}}, r_{\text{dual}}, r_{\text{comp}}\} = \max\{r_{\text{in}}, \mu^{(k)}\}. \tag{17.12}$$

The residual components and scaling convention are those of Chapter 15. The aggregate sum $-\boldsymbol{\nu}^{\mathsf{T}}\boldsymbol{h} = p\mu$, the infinity-norm residual μ, and the Euclidean residual $\sqrt{p}\,\mu$ are different quantities; they should not be interchanged in a stopping rule.

Algorithm: Practical barrier method

Choose a starting guess $\boldsymbol{z}^{(0)}$, $\mu^{(0)} > 0$, a fixed reduction factor $0 < \theta < 1$, fixed inner tolerances $\varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} > 0$, and $\varepsilon_{\text{out}} > \varepsilon_{\text{in}}^{(0)}$.

1. If the starting guess is not strictly inequality-feasible, run Phase I. Return a failure report if it does not supply $\boldsymbol{x}^{(0)} \in \mathcal{D}_<$. Initialize an equality-multiplier estimate $\boldsymbol{\lambda}^{(0)}$, if needed.
2. For $k = 0, 1, \ldots$, keep $\mu^{(k)}$ fixed and warm-start the subproblem from the preceding point and multiplier. Use an appropriate inner solver with actual-domain tests and sufficient-decrease backtracking until $r_{\text{in}} \le \varepsilon_{\text{in}}^{(k)}$.
3. Set $\nu_j^{(k+1)} = \mu^{(k)}/[-h_j(\boldsymbol{x}^{(k+1)})]$. Stop if all original KKT residuals satisfy $r_{\text{KKT}} \le \varepsilon_{\text{out}}$.
4. Otherwise set $\mu^{(k+1)} = \theta\mu^{(k)}$, and repeat the inner solve. If an inner solve fails, report that failure instead of reducing μ as though the subproblem had been solved.

For a finite-accuracy computation, a fixed $\varepsilon_{\text{in}}^{(k)} \equiv \varepsilon_{\text{in}}^{(0)} < \varepsilon_{\text{out}}$ is sufficient for the residual target (17.12), provided each inner solve succeeds. The accompanying code uses this fixed-tolerance version and a fixed reduction factor; its outer routine is `barrier_method`, initialization is `find_strict_start`, and the Newton inner solver is `newton_subproblem`. For an asymptotic argument, taking $\varepsilon_{\text{in}}^{(k)} \to 0$ guarantees vanishing inner errors as $\mu^{(k)} \to 0$; a fixed tolerance alone does not. Reducing μ too aggressively can move the next solution outside the region in which the previous point is a useful Newton start.

17.4 Comparison on a common nonlinear inequality problem

17.4.1 The test problem

We now compare the basic barrier method, inequality QP, and inequality ALM on the same two-dimensional problem. Retain the objective of Chapter 16 and replace its equality by an inequality:

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \frac{1}{2}[(x_1-1)^2+(x_2-2)^2] \qquad \text{subject to} \qquad h(\boldsymbol{x}) = x_1^2 + x_2 - 1 \le 0. \tag{17.13}$$

The objective is strictly convex, the constraint is convex, and the point $(7/32, 7/16)$ returned by Phase I is strictly feasible. The unconstrained minimizer $(1, 2)$ violates the inequality, so the constrained minimizer lies on its boundary. We can obtain a reference solution directly from the KKT equations. Stationarity is

$$x_1 - 1 + 2\nu x_1 = 0, \qquad x_2 - 2 + \nu = 0,$$

which, for $\nu \ge 0$, gives

$$x_1(\nu) = \frac{1}{1+2\nu}, \qquad x_2(\nu) = 2-\nu, \qquad s(\nu) := -h(\boldsymbol{x}(\nu)) = \nu - 1 - \frac{1}{(1+2\nu)^2}. \tag{17.14}$$

Because $s'(\nu) = 1+4/(1+2\nu)^3 > 0$, the active-boundary equation $s(\nu^*) = 0$ has one positive root. It gives

$$\boldsymbol{x}^* \approx (0.312908409,\, 0.902088327), \qquad \nu^* \approx 1.097911673.$$

Convex KKT sufficiency from Chapter 14 confirms global optimality, and strict convexity gives uniqueness. The solution therefore agrees with the equality solution of Chapter 16, but feasibility is now one-sided.

We can first describe the exact barrier and QP subproblem paths without running an inner solver. The barrier relation is $\mu = \nu s(\nu)$, with $\nu > \nu^*$ and positive slack. On this interval,

$$\frac{d}{d\nu}[\nu s(\nu)] = s(\nu) + \nu s'(\nu) > 0,$$

so each $\mu > 0$ gives one path point. For QP, $\nu = \rho[h]_+$. The stationary point must have $h > 0$, since otherwise it would be the infeasible unconstrained minimizer. Substitution into (17.14) gives

$$s(\nu) = -\nu/\rho,$$

with a unique root $0 < \nu < \nu^*$. Thus the barrier and QP again approach the same boundary from opposite sides. These scalar equations provide reference paths for checking the numerical results; the code solves the two-dimensional subproblems with Newton steps.

For ALM, use the inequality extension derived in Chapter 15:

$$A_\rho(\boldsymbol{x}; \nu^{(k)}) = f(\boldsymbol{x}) + \frac{[\nu^{(k)} + \rho h(\boldsymbol{x})]_+^2 - (\nu^{(k)})^2}{2\rho}, \qquad \nu^{(k+1)} = [\nu^{(k)} + \rho h(\boldsymbol{x}^{(k+1)})]_+.$$

The multiplier update is performed after every successful inner solve, with a fixed ρ. Differentiating the subproblem identifies $\widehat{\nu} = [\nu^{(k)} + \rho h]_+$, so its stationarity equation is again original stationarity with $\widehat{\nu}$. On this test, an exact subproblem solution cannot have $\widehat{\nu} = 0$, because that would force the infeasible unconstrained minimizer, where the projected coefficient is positive. Consequently (17.14) yields

$$\nu^{(k+1)} + \rho s(\nu^{(k+1)}) = \nu^{(k)}.$$

Starting at $\nu^{(0)} = 0$, suppose $\nu^{(k)} < \nu^*$. The left-hand side is strictly increasing; at $\nu^{(k)}$ it is smaller than the right-hand side, whereas at ν^* it is larger. Hence $\nu^{(k)} < \nu^{(k+1)} < \nu^*$. The exact ALM minimizers also have negative $s(\nu^{(k+1)})$, so they lie outside. The increasing, bounded multiplier sequence has a limit; passing to the limit in its update gives $s(\bar{\nu}) = 0$, hence $\bar{\nu} = \nu^*$. Thus the minimizers approach the boundary while multiplier corrections allow ρ to remain fixed. Differentiating the scalar update at ν^* gives the local contraction factor $1/[1 + \rho s'(\nu^*)]$, which is less than one.

17.4.2 Smaller parameter error can mean harder inner solves

To quantify the conditioning issue, write

$$\boldsymbol{a} = (2x_1, 1)^\mathsf{T}, \qquad \boldsymbol{D}(\nu) = \begin{pmatrix} 1 + 2\nu & 0 \\ 0 & 1 \end{pmatrix}.$$

Along the exact barrier path, (17.9) gives

$$\nabla^2 \Phi_\mu = \boldsymbol{D}(\nu) + \frac{\nu^2}{\mu} \boldsymbol{a}\boldsymbol{a}^\mathsf{T}. \tag{17.15}$$

The first term stays uniformly positive definite and bounded as $\nu \to \nu^*$. The rank-one second term grows in the normal direction $\boldsymbol{a}$, so its largest eigenvalue is $\Theta(1/\mu)$. To check the smaller eigenvalue, take the tangent vector $\boldsymbol{z} = (1, -2x_1)^\mathsf{T}$, for which $\boldsymbol{a}^\mathsf{T}\boldsymbol{z} = 0$. Its Rayleigh quotient is

$$\frac{\boldsymbol{z}^\mathsf{T} \nabla^2 \Phi_\mu \boldsymbol{z}}{\boldsymbol{z}^\mathsf{T}\boldsymbol{z}} = \frac{1 + 2\nu + 4x_1^2}{1 + 4x_1^2},$$

which remains bounded. Since $\nabla^2 \Phi_\mu \succeq \boldsymbol{I}$, the smallest eigenvalue is also bounded away from zero. Consequently $\kappa_2(\nabla^2 \Phi_\mu) = \Theta(1/\mu)$.

On the active QP branch, differentiating $P_\rho = f + \frac{\rho}{2}[h]_+^2$ instead gives

$$\nabla^2 P_\rho = \boldsymbol{D}(\nu) + \rho \boldsymbol{a}\boldsymbol{a}^\mathsf{T}, \qquad \nu = \rho h > 0.$$

The same tangent-normal argument yields $\kappa_2(\nabla^2 P_\rho) = \Theta(\rho)$ along its path. See Figure 17.6 for a comparison. On a positive ALM branch, the Hessian has the same form $\boldsymbol{D}(\widehat{\nu}) + \rho \boldsymbol{a}\boldsymbol{a}^\mathsf{T}$, but ρ is fixed and $\widehat{\nu} \to \nu^*$, so this penalty-induced growth is absent along the convergent sequence. Unlike the scalar example, the barrier and QP calculations demonstrate an actual growing ratio of curvatures. Both Hessians are positive definite on their smooth branches for this test. At $h = 0$, QP is continuously differentiable but has a jump in its Hessian, so its implementation uses the current branch and verifies progress by Armijo backtracking.

17.4.3 Numerical comparisons

Use the shared strict start $\boldsymbol{x}^{(0)} = (7/32, 7/16)$ obtained by `find_strict_start` above. All three algorithms use this point, $\varepsilon_{\text{out}} = 10^{-6}$, and fixed inner tolerances $\varepsilon_{\text{in}}^{(k)} \equiv 10^{-7}$. The barrier starts at $\mu^{(0)} = 1$ and multiplies it by 0.2 after an unsuccessful outer stopping test. QP starts at $\rho^{(0)} = 1$ and multiplies it by 5. ALM uses $\rho = 5$, starts with $\nu^{(0)} = 0$, and performs the projected multiplier update after every inner solve.

All methods warm-start the next subproblem and use Newton directions with Armijo constants $\sigma = 10^{-4}$, $\beta = 1/2$. The barrier also uses $\tau = 0.99$ for its linear fraction-to-boundary

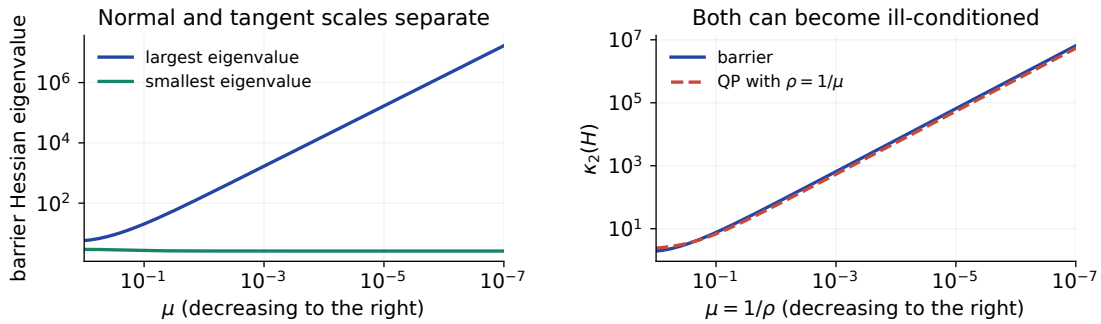


Figure 17.6: Conditioning along the exact paths for (17.13). Reducing the barrier parameter and increasing the penalty parameter both increase normal curvature relative to tangent curvature.

prediction and tests actual strict inequality feasibility before evaluating any logarithm. The table counts only outer subproblem solves and their Newton steps. The two Phase I SD steps are reported separately: the barrier requires an interior start, while QP and ALM could have started at the original infeasible guess. Sharing the Phase I output makes the optimization comparison use identical primal starting data.

	Barrier	QP	ALM
Outer solves / Newton steps	10 / 82	10 / 33	8 / 19
Stationarity r_{stat}	9.511×10^{-12}	2.614×10^{-10}	6.340×10^{-12}
Primal feasibility r_{feas}	0	5.621×10^{-7}	3.307×10^{-7}
Dual feasibility r_{dual}	0	0	0
Complementarity r_{comp}	5.120×10^{-7}	6.172×10^{-7}	3.631×10^{-7}
Overall r_{KKT}	5.120×10^{-7}	6.172×10^{-7}	3.631×10^{-7}

The barrier returns

$$\boldsymbol{x} \approx (0.312908328,\ 0.902087912), \qquad \nu \approx 1.097912088, \qquad \mu = 5.12 \times 10^{-7},$$

while QP returns

$$\boldsymbol{x} \approx (0.312908508,\ 0.902088828), \qquad \nu \approx 1.097911172, \qquad \rho = 1\,953\,125,$$

and ALM returns

$$\boldsymbol{x} \approx (0.312908467,\ 0.902088622), \qquad \nu \approx 1.097911378,$$

using the same fixed $\rho = 5$. Complementarity is the largest final residual in each run. Every Newton step solves a 2×2 system. QP and ALM need fewer such solves on this test, while the barrier guarantees strict inequality feasibility after initialization, including at every accepted inner iterate. These are different priorities: a barrier iterate stays inside, whereas QP and ALM work toward feasibility. The counts do not establish a general speed ranking; they depend on the problem, parameter factors, starting point, and inner accuracy. The convergence of KKT violations and the trajectories of primal variables for the three algorithms are plotted in Figure 17.7.

The outer routines are `barrier_method` for the barrier method, `quadratic_penalty` for QP, and `augmented_lagrangian` for ALM. The function `run_comparison` runs them on (17.13).

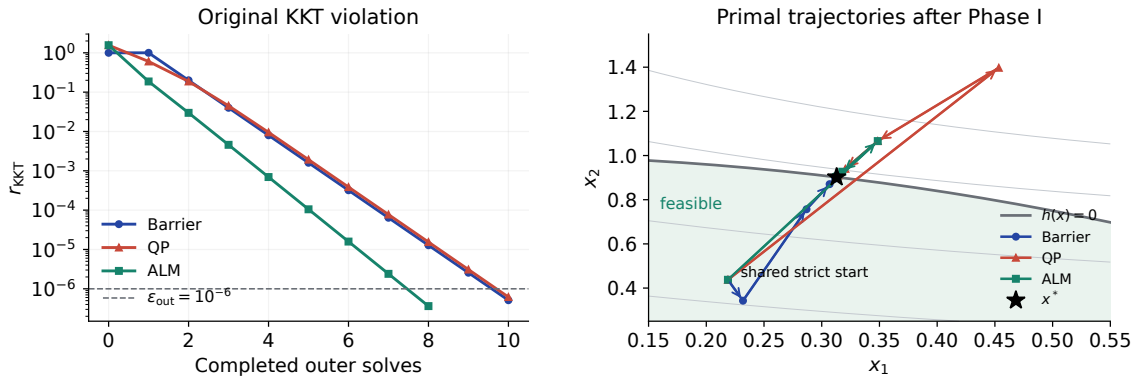


Figure 17.7: Barrier, QP, and ALM on the common nonlinear inequality problem. KKT violation measures all original conditions, including complementarity; the barrier's zero feasibility violation alone would not measure convergence. The curves begin at the common initial point and then show the outputs of completed inner solves.

Their inner routines and stopping tests are shared. For this quadratic f and h, the code evaluates objective differences using the exact identities, with $\boldsymbol{v} = \boldsymbol{x}_{\text{trial}} - \boldsymbol{x}$,

$$\Delta f = \nabla f(\boldsymbol{x})^{\mathsf{T}}\boldsymbol{v} + \tfrac{1}{2}\|\boldsymbol{v}\|^2, \qquad \Delta h = \nabla h(\boldsymbol{x})^{\mathsf{T}}\boldsymbol{v} + v_1^2.$$

For a strict interior trial, the barrier change is then

$$\Delta\Phi_\mu = \Delta f - \mu \log\left(1 + \frac{\Delta h}{h(\boldsymbol{x})}\right).$$

Evaluating the last logarithm with `log1p` avoids subtracting two nearly equal logarithms. For the ALM term, write $a = [\nu^{(k)} + \rho h(\boldsymbol{x})]_+$ and $b = [\nu^{(k)} + \rho(h(\boldsymbol{x}) + \Delta h)]_+$; its change is $(b-a)(b+a)/(2\rho)$. This expression also handles a branch crossing. These identities improve the numerical evaluation of the same Armijo condition; they do not change the algorithm or its parameter rules. For the barrier, the actual trial constraint is still checked first.

17.5 Further theory and extensions

The computations above require only the barrier equations and their derivatives. We now supply the regularity arguments that explain their local behavior and clarify what is, and is not, implied by a convergent primal sequence.

For the asymptotic results below, require vanishing inner errors. One possible schedule takes $\varepsilon_{\text{in}}^{(k)}$ proportional to $\mu^{(k)}$, which can avoid unnecessarily accurate early solves.

17.5.1 Convergence of induced multipliers

We will derive the missing multiplier limit under LICQ by separating active and inactive constraints at the primal limit.

Theorem 17.3 (KKT limit of inexact barrier points)**.** *Suppose $f, \boldsymbol{g}, \boldsymbol{h}$ are continuously differentiable near $\boldsymbol{x}^* \in \mathcal{D}$, $\mu^{(k)} > 0$ decreases to zero, and $\boldsymbol{x}^{(k)} \in \mathcal{D}_<$ converges to $\boldsymbol{x}^*$.*

Define $\nu_j^{(k)} = \mu^{(k)}/[-h_j(\boldsymbol{x}^{(k)})]$*, and suppose the equality-multiplier estimates* $\boldsymbol{\lambda}^{(k)}$ *satisfy*

$$\boldsymbol{g}(\boldsymbol{x}^{(k)}) \to \boldsymbol{0}, \qquad \nabla f(\boldsymbol{x}^{(k)}) + \boldsymbol{A}(\boldsymbol{x}^{(k)})\boldsymbol{\lambda}^{(k)} + \boldsymbol{C}(\boldsymbol{x}^{(k)})\boldsymbol{\nu}^{(k)} \to \boldsymbol{0}.$$

If LICQ holds at $\boldsymbol{x}^*$*, then* $(\boldsymbol{\lambda}^{(k)}, \boldsymbol{\nu}^{(k)})$ *converges to the unique multiplier pair* $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ *for which* $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ *satisfies the original KKT conditions.*

Proof. Continuity and the vanishing equality residual give $\boldsymbol{g}(\boldsymbol{x}^*) = \boldsymbol{0}$. The limit of strict inequality-feasible points gives $\boldsymbol{h}(\boldsymbol{x}^*) \le \boldsymbol{0}$. Let $I = \{j : h_j(\boldsymbol{x}^*) = 0\}$. For $j \notin I$, the slack is bounded away from zero for large k, and hence

$$\nu_j^{(k)} = \frac{\mu^{(k)}}{-h_j(\boldsymbol{x}^{(k)})} \longrightarrow 0.$$

Form $\boldsymbol{M}(\boldsymbol{x}) = [\boldsymbol{A}(\boldsymbol{x}), \boldsymbol{C}_I(\boldsymbol{x})]$. LICQ makes this matrix full column rank at $\boldsymbol{x}^*$ and, by continuity, at nearby iterates. Its left inverse $(\boldsymbol{M}^{\mathsf{T}}\boldsymbol{M})^{-1}\boldsymbol{M}^{\mathsf{T}}$ therefore converges. Writing the stationarity residual as $\boldsymbol{r}^{(k)} \to \boldsymbol{0}$, we have

$$\boldsymbol{M}(\boldsymbol{x}^{(k)}) \begin{pmatrix} \boldsymbol{\lambda}^{(k)} \\ \boldsymbol{\nu}_I^{(k)} \end{pmatrix} = -\nabla f(\boldsymbol{x}^{(k)}) - \sum_{j \notin I} \nu_j^{(k)} \nabla h_j(\boldsymbol{x}^{(k)}) + \boldsymbol{r}^{(k)}.$$

Multiplication by the left inverse proves convergence of the equality and active inequality multipliers. The inactive multipliers have already been shown to tend to zero. Positivity of $\boldsymbol{\nu}^{(k)}$ gives $\boldsymbol{\nu}^* \ge \boldsymbol{0}$. Finally, pass to the limit in stationarity and in $-\nu_j^{(k)} h_j(\boldsymbol{x}^{(k)}) = \mu^{(k)}$ to obtain stationarity and complementarity. LICQ makes the limiting multipliers unique. □

17.5.2 A local smooth central path

We next ask whether a regular KKT solution has nearby barrier solutions. The tool will be the implicit-function theorem, applied to the perturbed KKT equations with explicit slacks. We first write this system, then state the regularity conditions needed to obtain a local branch of its solutions.

Introduce $\boldsymbol{s} = -\boldsymbol{h}(\boldsymbol{x})$ and write the equations as

$$\nabla f(\boldsymbol{x}) + \boldsymbol{A}(\boldsymbol{x})\boldsymbol{\lambda} + \boldsymbol{C}(\boldsymbol{x})\boldsymbol{\nu} = \boldsymbol{0}, \tag{17.16}$$

$$\boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{0}, \tag{17.17}$$

$$\boldsymbol{h}(\boldsymbol{x}) + \boldsymbol{s} = \boldsymbol{0}, \tag{17.18}$$

$$\operatorname{diag}(\boldsymbol{s})\boldsymbol{\nu} = \mu \boldsymbol{1}. \tag{17.19}$$

The following assumptions are sufficient locally and are stronger than those needed merely to pass to a KKT limit.

Theorem 17.4 (Local smooth central path)**.** *Suppose* $f, \boldsymbol{g}, \boldsymbol{h}$ *are twice continuously differentiable near a KKT point* $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ *of* (17.1) *with* $\boldsymbol{x}^* \in \mathcal{D}$*. Assume LICQ, strict complementarity, and the second-order sufficient condition from Chapter 13. Then there is a* $\mu_0 > 0$ *and a locally unique continuously differentiable branch* $(\boldsymbol{x}(\mu), \boldsymbol{\lambda}(\mu), \boldsymbol{\nu}(\mu), \boldsymbol{s}(\mu))$*,* $0 \le \mu < \mu_0$*, solving* (17.16)–(17.19) *and taking the value* $(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*, -\boldsymbol{h}(\boldsymbol{x}^*))$ *at* $\mu = 0$*. For* $0 < \mu < \mu_0$*, the slacks and inequality multipliers are strictly positive, and* $\boldsymbol{x}(\mu)$ *is a strict local minimizer of the barrier subproblem.*

Proof. At $\mu = 0$, the KKT solution with $\boldsymbol{s}^* = -\boldsymbol{h}(\boldsymbol{x}^*)$ solves this system. We will first check that its Jacobian is nonsingular, then determine the signs along the resulting local branch, and finally establish that the branch consists of local minimizers.

Let $I = \{j : h_j(\boldsymbol{x}^*) = 0\}$. Define the residual in Equations (17.16) to (17.19) as a function of $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}, \boldsymbol{s}; \mu)$. At $\mu = 0$, its Jacobian with respect to $(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}, \boldsymbol{s})$ is

$$\boldsymbol{K}_* = \begin{pmatrix} \boldsymbol{H}_* & \boldsymbol{A}_* & \boldsymbol{C}_* & \boldsymbol{0} \\ \boldsymbol{A}_*^\mathsf{T} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{C}_*^\mathsf{T} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} \\ \boldsymbol{0} & \boldsymbol{0} & \operatorname{diag}(\boldsymbol{s}^*) & \operatorname{diag}(\boldsymbol{\nu}^*) \end{pmatrix},$$

where $\boldsymbol{H}_* = \nabla^2_{\boldsymbol{xx}} L(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ and L is the ordinary Lagrangian defined above.

We show that $\boldsymbol{K}_*$ is nonsingular. Let $(\boldsymbol{d}, \boldsymbol{\delta}, \boldsymbol{\eta}, \boldsymbol{\zeta})$ lie in its nullspace. For $j \notin I$, $s_j^* > 0$ and $\nu_j^* = 0$, so the fourth block equation gives $\eta_j = 0$. For $j \in I$, strict complementarity gives $\nu_j^* > 0$ and $s_j^* = 0$, so it gives $\zeta_j = 0$. The third block equation then implies $(\boldsymbol{C}_I^*)^\mathsf{T}\boldsymbol{d} = \boldsymbol{0}$, while the second gives $\boldsymbol{A}_*^\mathsf{T}\boldsymbol{d} = \boldsymbol{0}$. Multiplying the first block equation by $\boldsymbol{d}^\mathsf{T}$ eliminates the multiplier terms and gives $\boldsymbol{d}^\mathsf{T}\boldsymbol{H}_*\boldsymbol{d} = 0$. Under strict complementarity, the directions satisfying the two displayed orthogonality conditions form the critical subspace. SOSC therefore gives $\boldsymbol{d} = \boldsymbol{0}$. The third block now gives $\boldsymbol{\zeta} = \boldsymbol{0}$; the first and LICQ give $\boldsymbol{\delta} = \boldsymbol{0}$ and $\boldsymbol{\eta}_I = \boldsymbol{0}$, while we already know $\boldsymbol{\eta}_{I^c} = \boldsymbol{0}$. Thus the Jacobian is nonsingular.

The implicit-function theorem now gives a $\mu_0 > 0$ and unique continuously differentiable functions

$$(\boldsymbol{x}(\mu), \boldsymbol{\lambda}(\mu), \boldsymbol{\nu}(\mu), \boldsymbol{s}(\mu)), \qquad 0 \le \mu < \mu_0,$$

solving the displayed equations in a neighborhood of the original KKT triple and slack. This uniqueness concerns the local branch, not all barrier stationary points elsewhere in the domain. For active indices, $\nu_j(\mu)$ stays positive, so Equation (17.19) gives $s_j(\mu) = \mu/\nu_j(\mu) > 0$. For inactive indices, $s_j(\mu)$ stays positive, so $\nu_j(\mu) = \mu/s_j(\mu) > 0$. Eliminating $\boldsymbol{s}$ and $\boldsymbol{\nu}$ recovers the barrier KKT equations.

It remains to prove local minimality. On the equality tangent space, the barrier Lagrangian Hessian is

$$\boldsymbol{H}(\mu) + \boldsymbol{C}(\mu) \operatorname{diag}\left(\frac{\nu_j(\mu)}{s_j(\mu)}\right) \boldsymbol{C}(\mu)^\mathsf{T}, \tag{17.20}$$

where $\boldsymbol{H}(\mu) = \nabla^2_{\boldsymbol{xx}} L(\boldsymbol{x}(\mu), \boldsymbol{\lambda}(\mu), \boldsymbol{\nu}(\mu))$. Suppose this matrix were not positive definite on the equality tangent space along a sequence $\mu^{(k)} \downarrow 0$. Choose corresponding unit tangent directions $\boldsymbol{d}^{(k)}$ with nonpositive quadratic form and pass to a convergent subsequence. For $j \in I$, $\nu_j(\mu)/s_j(\mu) = \nu_j(\mu)^2/\mu \to \infty$. Since $\boldsymbol{H}(\mu)$ is bounded and all barrier-curvature terms are nonnegative, this forces $\nabla h_j(\boldsymbol{x}(\mu^{(k)}))^\mathsf{T}\boldsymbol{d}^{(k)} \to 0$ for every $j \in I$. The limiting unit direction is therefore in the critical subspace. The nonpositive total quadratic form also implies that its $\boldsymbol{H}(\mu^{(k)})$-part is nonpositive; passing to the limit contradicts SOSC. Thus Equation (17.20) is positive definite on the equality tangent space for small $\mu > 0$. The equality-constrained second-order sufficient condition proves strict local minimality for all sufficiently small positive μ, completing the proof. □

17.5.3 Extension to primal–dual interior-point methods

The barrier method above is a *primal interior-point method* (primal IPM): it substitutes $\nu_j = \mu/(-h_j)$ at every point. To extend it to a *primal–dual interior-point method* (primal–dual IPM), retain slacks $\boldsymbol{s} > \boldsymbol{0}$ and multipliers $\boldsymbol{\nu} > \boldsymbol{0}$ as separate variables in (17.16)–(17.19).

Define

$$\boldsymbol{r}_d = \nabla f + \boldsymbol{A}\boldsymbol{\lambda} + \boldsymbol{C}\boldsymbol{\nu}, \quad \boldsymbol{r}_g = \boldsymbol{g}, \quad \boldsymbol{r}_h = \boldsymbol{h} + \boldsymbol{s}, \quad \boldsymbol{r}_c = \operatorname{diag}(\boldsymbol{s})\boldsymbol{\nu} - \mu\mathbf{1}.$$

Newton linearization of these equations gives

$$\begin{pmatrix} \boldsymbol{H} & \boldsymbol{A} & \boldsymbol{C} & \mathbf{0} \\ \boldsymbol{A}^{\mathsf{T}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \boldsymbol{C}^{\mathsf{T}} & \mathbf{0} & \mathbf{0} & \boldsymbol{I} \\ \mathbf{0} & \mathbf{0} & \operatorname{diag}(\boldsymbol{s}) & \operatorname{diag}(\boldsymbol{\nu}) \end{pmatrix} \begin{pmatrix} \boldsymbol{d} \\ \boldsymbol{\delta} \\ \boldsymbol{\eta} \\ \boldsymbol{\zeta} \end{pmatrix} = - \begin{pmatrix} \boldsymbol{r}_d \\ \boldsymbol{r}_g \\ \boldsymbol{r}_h \\ \boldsymbol{r}_c \end{pmatrix}, \tag{17.21}$$

where $\boldsymbol{H} = \nabla^2_{\boldsymbol{x}\boldsymbol{x}} L$. The third row gives $\boldsymbol{\zeta} = -\boldsymbol{r}_h - \boldsymbol{C}^{\mathsf{T}}\boldsymbol{d}$; substituting into the fourth gives

$$\boldsymbol{\eta} = \operatorname{diag}(\boldsymbol{s})^{-1}\left[-\boldsymbol{r}_c + \operatorname{diag}(\boldsymbol{\nu})\boldsymbol{r}_h + \operatorname{diag}(\boldsymbol{\nu})\boldsymbol{C}^{\mathsf{T}}\boldsymbol{d}\right].$$

Insert this expression into the first row to obtain

$$[\boldsymbol{H} + \boldsymbol{C}\operatorname{diag}(\boldsymbol{\nu})\operatorname{diag}(\boldsymbol{s})^{-1}\boldsymbol{C}^{\mathsf{T}}]\boldsymbol{d} + \boldsymbol{A}\boldsymbol{\delta} = -\boldsymbol{r}_d + \boldsymbol{C}\operatorname{diag}(\boldsymbol{s})^{-1}\boldsymbol{r}_c - \boldsymbol{C}\operatorname{diag}(\boldsymbol{s})^{-1}\operatorname{diag}(\boldsymbol{\nu})\boldsymbol{r}_h,$$
$$\boldsymbol{A}^{\mathsf{T}}\boldsymbol{d} = -\boldsymbol{r}_g.$$

When $\boldsymbol{r}_h = \boldsymbol{r}_c = \mathbf{0}$, we have $\nu_j/s_j = \mu/h_j^2$, and this is exactly (17.10). Away from those equations, a primal–dual method can instead update the separate slacks and multipliers while reducing their residuals. Positivity of $\boldsymbol{s}$ does not imply $\boldsymbol{h}(\boldsymbol{x}) < \mathbf{0}$ unless $\boldsymbol{h}+\boldsymbol{s} = \mathbf{0}$; this is why a primal–dual method can start infeasibly in the original inequalities. Its step selection and globalization require additional machinery. The accompanying code implements the basic barrier method only.

17.6 Summary and comparison with previous methods

The roles of the methods from the last three chapters can be summarized as follows.

Method	How constraints enter	What needs attention
QP	Penalty for violations; ρ increases	Finite-parameter infeasibility; large normal curvature
ALM	Penalty plus multiplier correction	Multiplier accuracy and a sufficiently large fixed ρ in the local theory
LN	Newton equations for the original KKT system	Local initialization, nonsingular KKT matrix, and no automatic minimization safeguard
Equality SQP	Linearized constraints and a quadratic local model	Feasibility of the linearization, reduced curvature, and merit descent
Barrier	Logarithms on the strict inequality domain; μ decreases	An interior start, boundary step lengths, and inner accuracy

The chapters introduced QP, ALM, LN, and SQP mainly for equalities. Their inequality extensions require extra structure; the PHR form used above is the classical ALM extension. For a smooth equality subproblem, LN is the Newton root-finding view, while SQP supplies the optimization model and merit safeguards. For a mixed problem, equality SQP can also be the inner solver of a barrier method. There is therefore no conflict between these viewpoints and no universal speed ordering. The choice depends on constraint structure, the availability of an interior point, and the cost of the inner linear algebra. SD, Newton, and the other unconstrained methods studied earlier remain available as inner solvers; their

inner stopping criteria must be related to the original KKT residual. For every method, record inner work as well as outer iterations and check the actual conditions intended by the algorithm: strict inequality feasibility for the barrier method, linearized feasibility for an SQP step, and original KKT feasibility and stationarity at termination.

17.7 Exercises

Exercise 17.1. For $\min_x x$ subject to $x^2 \leq 1$, identify the strict interior, write the log-barrier objective, solve its stationarity equation, select the interior branch, and compute the induced multiplier and its limit. Verify perturbed complementarity before taking the limit.

Exercise 17.2. For the mixed problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} -x_1 \quad \text{subject to} \quad x_1 + x_2 = 0, \qquad x_1 - 1 \leq 0,$$

derive the exact central path and both multipliers. Verify the KKT limit and compute the barrier-Hessian curvature in the active normal direction.

Exercise 17.3. For affine inequalities $\boldsymbol{C}^\mathsf{T}\boldsymbol{x} - \boldsymbol{b} < \boldsymbol{0}$, write the barrier Hessian in matrix form. Identify the directions receiving positive barrier curvature. Derive the fraction-to-boundary step bound for a given direction, and explain why the same linear bound alone is insufficient for nonlinear inequalities.

Exercise 17.4. Construct a problem with a nonempty feasible set but empty strict interior. Explain why the basic logarithmic-barrier formulation is unavailable and suggest a reformulation or another method from Chapters 15 and 16.